Relations and Functions

- **Definition 1**
  Let X be a set. A **relation** $\sim$ on X is a rule that determines for every $x, y \in X$ whether the statement

  $$x \sim y$$

  is true or false. If it is true, we say that x is related to y. If it is false, we write $x \nsim y$.

  - **Example 1.1**
    Let $X = \{1, 2, 3, 4\}$, the following are examples of relations on X.
    1. The relation $\sim$ given by $1 \sim 2, 2 \sim 3, 3 \sim 1, 4 \sim 4$.
    2. The relation $\sim$ given by $1 \sim 1, 2 \sim 2, 2 \sim 4$.
    3. The relation $\sim$ given by $1 \sim 1, 2 \sim 2, 3 \sim 3, 4 \sim 4$.

- **Definition 2**
  Let X be a set and $\sim$ a relation on X.
  1. The relation is **reflexive** if $x \sim x$ for every $x \in X$.
  2. The relation is **symmetric** if for every $x, y \in X$

     $$x \sim y \implies y \sim x.$$

  3. The relation is **antisymmetric** if for every $x, y \in X$

     $$x \sim y \wedge y \sim x \implies x = y.$$

  4. The relation is **transitive** if for every $x, y, z \in X$

     $$x \sim y \wedge y \sim z \implies x \sim z.$$

  - **Example 2.1**
    On the set $\mathbb{Z}$ consider the relation

    $$x \sim y \iff x \leqslant y.$$

    Note that
    1. $\sim$ is reflexive since $x \leqslant x$ for every $x \in \mathbb{Z}$.
    2. $\sim$ is not symmetric since $1 \leqslant 2$ but $2 \nleqslant 1$.
    3. $\sim$ is antisymmetric because if $x \leqslant y$ and $y \leqslant x$ then $x = y$.
    4. $\sim$ is transitive since $x \leqslant y$ and $y \leqslant z$ implies that $x \leqslant z$.

- **Definition 3**
  Let X be a set. An **equivalence relation** on X is a relation $\sim$ that satisfies the following properties.
  1. $\sim$ is reflexive.
  2. $\sim$ is symmetric.
  3. $\sim$ is transitive.

  - **Example 3.1**
    Let $X = \{1, 2, 3, 4\}$. The following are examples of equivalence relations on X.
    1. The relation $\sim$ given by $1 \sim 1, 2 \sim 2, 3 \sim 3, 4 \sim 4$.

2. The relation $\sim$ given by $1 \sim 1, 2 \sim 2, 3 \sim 3, 4 \sim 4, 2 \sim 3, 3 \sim 2, 1 \sim 4, 4 \sim 1.$

- **Definition 4**
  Let X be a set. An **order relation** on X is a relation $\sim$ that satisfies the following properties.
  1. $\sim$ is reflexive.
  2. $\sim$ is antisymmetric.
  3. $\sim$ is transitive.
  - **Example 4.1**
    Let $X = \{1, 2, 3, 4\}$. The following are examples of order relations on X.
    1. The relation $\sim$ given by $1 \sim 1, 2 \sim 2, 3 \sim 3, 4 \sim 4.$
    2. The relation $\sim$ given by $1 \sim 1, 2 \sim 2, 3 \sim 3, 4 \sim 4, 1 \sim 2, 1 \sim 3, 1 \sim 4, 2 \sim 3, 2 \sim 4, 3 \sim 4.$

- **Definition 5**
  A **function** $f: X \to Y$ is a correspondence of elements of X with elements of Y such that forall $x \in X$ there exists a unique $y \in Y$ such that $y = f(x)$.

$$\begin{aligned} f: X \to &\ Y \\ x \mapsto &\ y = f(x) \end{aligned}$$

  The element $y = f(x)$ is called the **image** of x under f and the element x is called the **preimage** of y under f.

- **Definition 6**
  If $f: X \to Y$ is a function, the **domain** of f denote $\mathrm{Dom}(f)$ is X and the **codomain** of $\mathrm{CoDom}(f)$ is Y. The **range** of f, denoted $\mathrm{Ran}(f)$, is the set

$$\mathrm{Ran}(f) = \{y \in Y : y = f(x) \text{ for some } x \in X\}$$

  Note that $\mathrm{Ran}(f) \subseteq Y$.
  - **Example 6.1**
    1. The function $f: \mathbb{R} \to \mathbb{R}$, $f(x) = x^2$.
    2. The function $g: \mathbb{R} \to \mathbb{R}$, $g(x) = \sin(x)$.
    3. The function $h: [0, \infty) \to \mathbb{R}$, $h(x) = \sqrt{x}$.
  - **Example 6.2**
    Let $X = \{1, 2, 3\}$ and $Y = \{a, b, c\}$ the following examples of functions from X to Y.
    1. The function $f: X \to Y$ defined by

$$f(1) = a \quad f(2) = b \quad f(3) = c.$$

    2. The function $g: X \to Y$ defined by

$$g(1) = a \quad g(2) = a \quad g(3) = a.$$

    3. The function $h: X \to Y$ defined by

$$h(1) = a \quad h(2) = b \quad h(3) = a.$$

- **Definition 7**

Two functions $f: X \to Y$ and $g: Z \to W$ are **equal** if $X = Z, Y = W$ and $f(x) = g(x)$ for all $x \in X$.

- **Example 7.1**
  1. Let X be a set. The **identity function** on X is the function $\mathrm{id}_x: X \to X$ given by $\mathrm{id}_x(x) = x$.
  2. Let X and Y be sets and $c \in Y$. The **constant function** with value c is the function $\mathrm{K}_c: X \to Y$ given by $\mathrm{K}_c(x) = c$.
  3. Let X be a set and $A \subseteq X$. The **inclusion function** of A in X is the function $\iota: A \to X$ (or $\iota: A \hookrightarrow X$) given by $\forall x \in A, \iota(x) = x$.
  4. If $f: X \to Y$ is a function and $A \subseteq X$, the **restriction** of f to A is the function $f\big|_A: A \to Y$ given by $f\big|_A(x) = f(x)$ for all $x \in A$.

- **Definition 8**
  If $f: X \to Y$ and $g: Z \to W$ are functions, the **composition** of g with f, denoted $g \circ f$, is the function $g \circ f: X \to Z$ defined by $(g \circ f) = g\left(f\left(x\right)\right)$ for all $x \in X$.
  - **Example 8.1**
    Note that if $f: X \to Y$, f is a function and $A \subseteq X$, we have that $f\big|_A = f \circ \iota_A$ where $\iota: A \to x$.

$$A \xrightarrow{\iota} X \xrightarrow{f} Y.$$

- **Remark 8**
  Note that even in the case both $g \circ f$ and $f \circ g$ are defined, the composition of two functions is not commutative, that is, in general $g \circ f \neq f \circ g$.
  - **Example 8.2**
    Consider the functions

$$\begin{aligned} f\colon & \mathbb{R} \to \mathbb{R}, & g\colon & \mathbb{R} \to \mathbb{R}. \\ & x \mapsto x^2, & & x \mapsto x+1. \end{aligned}$$

    Note thet $g \circ f: \mathbb{R} \to \mathbb{R}$ and $f \circ g: \mathbb{R} \to \mathbb{R}$ with

$$\begin{aligned} g \circ f &= g\left(f\left(x\right)\right) & f \circ g &= f\left(g\left(x\right)\right) \\ &= g\left(x^2\right) & &= f\left(x+1\right) \\ &= x^2 + 1 & &= \left(x+1\right)^2. \end{aligned}$$

    In particular if $x = 2$ we have that $(g \circ f)(2) = 5$ and $(f \circ g) = 9$. Therefore $g \circ f \neq f \circ g$.
  - **Proposition 8.3**
    Let $f: X \to Y, g: Y \to Z$ and $h: Z \to W$ be functions, the $(h \circ g) \circ f = h \circ (g \circ f)$. In other words, the composition is associative.
    - **Proof 8.3.1**
      Let $x \in X$ then

$$\begin{aligned} [(h \circ g) \circ f](x) &= (h \circ g)(f(x)) \\ &= h(g(f(x))) \\ &= h((g \circ f)(x)) \\ &= [h \circ (g \circ f)](x). \end{aligned}$$

      Now, as $[(h \circ g) \circ f](x) = [h \circ (g \circ f)](x)$ for all $x \in X$ we have that $(h \circ g) \circ f = h \circ (g \circ f)$.
  - **Example 8.4**
    Let $s, t: \mathbb{Z} \to \mathbb{Z}$ be functions defined by $s(x) = x + 1$ and $t(x) = 2x$. Prove that $t \circ s \neq s \circ t$.

- **Proof 8.4.1**
  Let $x \in \mathbb{Z}$ then

$$\begin{aligned} t \circ s &= t(s(x)) & s \circ t &= s(t(x)) \\ &= t(x+1) & &= s(2x) \\ &= 2x+2 & &= 2x+1. \end{aligned}$$

  In particular if $x = 2$ we have that $(t \circ s)(2) = 6$ and $(s \circ t)(2) = 5$. Therefore $t \circ s \neq s \circ t$.

- **Example 8.5**
  Let $X = \{1, 2, 3, 4, 5\}$ and let $f\colon X \to X$ be the function defined by

$$f(1) = 2, \quad f(2) = 2, \quad f(3) = 4, \quad f(4) = 4, \quad f(5) = 4.$$

  Show that $f \circ f = f$. Find a function g such that $g \circ f = f$ and $f \circ g = f$.

  - **Solution 8.5.1**

$$g\colon X \to X$$
$$g(1) = 1, \quad g(2) = 2, \quad g(3) = 5, \quad g(4) = 4, \quad g(5) = 3.$$

  Then

$$\begin{aligned} (g \circ f)(1) = g(f(1)) = g(2) = 2 = f(1), \\ (g \circ f)(2) = g(f(2)) = g(2) = 2 = f(2), \\ (g \circ f)(3) = g(f(3)) = g(4) = 4 = f(3), \\ (g \circ f)(4) = g(f(4)) = g(4) = 4 = f(4), \\ (g \circ f)(5) = g(f(5)) = g(4) = 4 = f(5). \end{aligned}$$

  Therefore $g \circ f = f$. Similarly, we can prove $f \circ g = f$ and $f \circ f = f$.

- **Definition 9**
  Let $f\colon X \to Y$ be a function.
  We say that f is **injective** if given $x_1, x_2 \in X$, such that $f(x_1) = f(x_2)$ implies $x_1 = x_2$.
  Equivalently, if given $x_1, x_2 \in X$ such that $x_1 \neq x_2$ implies $f(x_1) \neq f(x_2)$.

  We say that f is **surjective** if for every $y \in Y$ there exists $x \in X$ such that $y = f(x)$.

  We say that f is **bijective** if it is both injective and surjective.

- **Remark 9**
  If there exists a bijective function $f\colon X \to Y$. We say that X and Y are **one-to-one correspondence**.

  - **Example 9.1**
    The identity function $\mathrm{id}_X\colon X \to X$ is injective since if $\mathrm{id}_X(x) = \mathrm{id}_X(y)$ then $x = y$.
    It is also surjective since for all $x \in X$ we have that $x = \mathrm{id}_X(x)$.
    Therefore, id_X is bijective.

  - **Example 9.2**
    Consider the constant function $\mathrm{K}_c\colon X \to Y$.
    Note that if X has more than one element the function is not injective and if Y has more than one element the function is not surjective.

  - **Example 9.3**
    The inclusion function $\iota\colon A \to X$ is injective since $\iota_A(x) = \iota_A(y)$ implies $x = y$. Note that, if $A = X$ then $\iota_x = \mathrm{id}_X$. Therefore ι_A is surjective. If $A \neq X$ and is not surjective if $A \subset X$.

  - **Example 9.4**
    Suppose that $f\colon A \to B$ and $g\colon B \to C$ are functions. Prove that if $g \circ f$ is injective then f is injective, and if $g \circ f$ is surjective then g is surjective.

- **Solution 9.4.1**
  Proof:
    - **First Part (If $g \circ f$ is injective, then f is injective):**
      Let $x_1, x_2 \in A$ be arbitrary elements in the domain of f such that:
      $$f(x_1) = f(x_2)$$
      Since g is a function, applying g to both sides yields identical outputs:
      $$g(f(x_1)) = g(f(x_2))$$
      By definition of function composition, this is equivalent to:
      $$(g \circ f)(x_1) = (g \circ f)(x_2)$$
      We are given the hypothesis that $g \circ f$ is injective. By definition of an injective function, $(g \circ f)(x_1) = (g \circ f)(x_2) \implies x_1 = x_2$.
      Thus, $f(x_1) = f(x_2) \implies x_1 = x_2$, proving that f is injective.
    - **Second Part (If $g \circ f$ is surjective, then g is surjective):**
      Let $z \in C$ be an arbitrary element in the codomain of g.
      We are given the hypothesis that $g \circ f \colon A \to C$ is surjective. By definition of a surjective function, there exists some element $x \in A$ such that:
      $$(g \circ f)(x) = z$$
      By definition of function composition, this means:
      $$g(f(x)) = z$$
      Let $y = f(x)$. Since $x \in A$ and $f \colon A \to B$, it follows that $y \in B$.
      Substituting y back into the equation gives $g(y) = z$.
      Thus, for any arbitrary $z \in C$, there exists an element $y \in B$ such that $g(y) = z$, proving that g is surjective.
- **Example 9.5**
  Prove that the function $f \colon \mathbb{R} \to \mathbb{R}$ defined by $f(x) = 2x + 1$ is injective and surjective.
    - **Proof 9.5.1**
      Injective Suppose that $f(x_1) = f(x_2)$, that is,
      $$\begin{aligned} 2x_1 + 1 &= 2x_2 + 1 \\ 2x_1 &= 2x_2 \\ x_1 &= x_2 \end{aligned}$$
      then f is injective.
      Surjective Let $y \in \mathbb{R}$ and consider $x = \dfrac{y-1}{2} \in \mathbb{R}$, then $f(x) = f\left(\dfrac{y-1}{2}\right) = 2\left(\dfrac{y-1}{2}\right) + 1 = y - 1 + 1 = y$.
      Therefore f is surjective.

- **Definition 10**
  Let $f \colon X \to Y$ be a function. If $A \subseteq X$ the **direct image** of A under f, denoted $f(A)$, is the set
  $$f(A) = \{y \in Y : y = f(x) \text{ for some } x \in A\}$$
  Note that $f(A) \subseteq Y$ abd $f(X) = \mathrm{Ran}(f)$.

If $B \subseteq Y$ the **inverse image** of B under f, denoted $f^{-1}(B)$, is the set

$$f^{-1}(B) = \{x \in X : f(x) \in B\}$$

- **Example 10.1**
  Let $X = \{1,2,3,4\}, Y = \{a,b,c,d\}$ and $f\colon X \to Y$ defined by $f(1) = a$, $f(2) = c$, $f(3) = c$, $f(4) = a$.
  Consider $A = \{1,2,4\} \subseteq X$ and $B = \{a,b\} \subseteq Y$. Then

  $$f(A) = \{a,c\}\,, \quad f^{-1}(B) = \{1,4\}$$
  $$f^{-1}(\{c\}) = \{2,3\}\,, \quad f^{-1}(\{d\}) = \varnothing$$

- **Proposition 10.2**
  Let $f\colon X \to Y$ a function, $A, B \subseteq X$ and $C, D \subseteq Y$.
  1. $f(A \cap B) \subseteq f(A) \cap f(B)$.
     Proof
     Let $y \in f(A \cap B)$ then $y = f(x)$ for some $x \in A \cap B$. Since $x \in A \cap B$ then $x \in A$ and $x \in B$, that is, $y = f(x) \in f(A)$ and $y = f(x) \in f(B)$. Hence $y \in f(A) \cap f(B)$ and thus $f(A \cap B) \subseteq f(A) \cap f(B)$.
  2. $f(A \cup B) = f(A) \cup f(B)$.
     Proof
     ($\subseteq$) Let $y \in f(A \cup B)$ then $y = f(x)$ for some $x \in A \cup B$. Since $x \in A \cup B$ then $x \in A$ or $x \in B$, that is, $y = f(x) \in f(A)$ or $y = f(x) \in f(B)$ hence $y \in f(A) \cup f(B)$ and thus $f(A \cup B) \subseteq f(A) \cup f(B)$.

     ($\supseteq$) Let $y \in f(A) \cup f(B)$ then $y \in f(A)$ or $y \in f(B)$, that is, $y = f(a)$ for some $a \in A$ or $y = f(b)$ for some $b \in B$. Therefore $y \in f(A \cup B)$ since $a, b \in A \cup B$ and thus $f(A) \cup f(B) \subseteq f(A \cup B)$.
  3. $f^{-1}(C \cap D) = f^{-1}(C) \cap f^{-1}(D)$.
     Proof
     ($\subseteq$) Let $x \in f^{-1}(C \cap D)$ then $f(x) \in C \cap D$, that is, $f(x) \in C$ and $f(x) \in D$ hence $x \in f^{-1}(C)$ and $x \in f^{-1}(D)$ therefore $x \in f^{-1}(C) \cap f^{-1}(D)$ and thus $f^{-1}(C \cap D) \subseteq f^{-1}(C) \cap f^{-1}(D)$.

     ($\supseteq$) Let $x \in f^{-1}(C) \cap f^{-1}(D)$ then $x \in f^{-1}(C)$ and $x \in f^{-1}(D)$, that is, $f(x) \in C$ and $f(x) \in D$ hence $f(x) \in C \cap D$, that is, $x \in f^{-1}(C \cap D)$ and thus $f^{-1}(C) \cap f^{-1}(D) \subseteq f^{-1}(C \cap D)$.
  4. $f^{-1}(C \cup D) = f^{-1}(C) \cup f^{-1}(D)$.
     Proof
     ($\subseteq$) Let $x \in f^{-1}(C \cup D)$ then $f(x) \in C \cup D$, that is, $f(x) \in C$ or $f(x) \in D$ hence $x \in f^{-1}(C)$ or $x \in f^{-1}(D)$ therefore $x \in f^{-1}(C) \cup f^{-1}(D)$ and thus $f^{-1}(C \cup D) \subseteq f^{-1}(C) \cup f^{-1}(D)$.

     ($\supseteq$) Let $x \in f^{-1}(C) \cup f^{-1}(D)$ then $x \in f^{-1}(C)$ or $x \in f^{-1}(D)$, that is, $f(x) \in C$ or $f(x) \in D$ hence $f(x) \in C \cup D$, that is, $x \in f^{-1}(C \cup D)$ and thus $f^{-1}(C) \cup f^{-1}(D) \subseteq f^{-1}(C \cup D)$.

- **Remark 10.2**
  Note that $f(A) \cap f(B) \subseteq f(A \cap B)$ is not always true, for example consider the function $f\colon \mathbb{R} \to \mathbb{R}$ given by $f(x) = x^2$, $A = \{-1\}$ and $B = \{1\}$ then $f(A) = 1 = f(B)$. But $f(A \cap B) = f(\varnothing) = \varnothing$.

- **Exercise 10.3**
  If $f\colon X \to Y$ is injective and $A, B \subseteq X$ then $f(A) \cap f(B) \subseteq f(A \cap B)$.

- **Theorem 10.4**

Let $f: X \to Y$ be an injective function. Let $y \in \operatorname{Ran}$ then there exists a **unique** $x \in X$ such that $y = f(x)$, that is, $f^{-1}(\{y\}) = \{x\}$.

- **Proof 10.4.1**
  Let $y \in \operatorname{Ran}(f)$ then there exists $x \in X$ such that $y = f(x)$. Suppose that there exists $x' \in X$ such that $y = f(x')$ then $f(x) = f(x')$ and as f is injective we have that $x = x'$.

- **Definition 11**
  A funtion $f: X \to Y$ is called **invertible** if there exists a function $g: Y \to X$ such that $g \circ f = \mathrm{id}_X$ and $f \circ g = \mathrm{id}_Y$. Such that function is called **inverse** of f.
- **Remark 11**
  If $f: X \to Y$ is invertible then its inverse is unique. In fact, suppose that $g, g': Y \to X$ are both inverse of f, then

$$g \circ f = \mathrm{id}_X = g' \circ f \text{ and } f \circ g = \mathrm{id}_Y = f \circ g'$$

hence $g = \mathrm{id}_X \circ g = (g' \circ f) \circ g = g' (\circ f \circ g) = g' \circ \mathrm{id}_Y = g'$.

If f is invertivle, the inverse of f is denoted f^{-1}. Note that for all $x \in X$ and $y \in Y$

$$y = f(x), \text{ if and only if, } f^{-1}(y) = x$$

- **Example 11.1**
  Let $f: \mathbb{R} \to \mathbb{R}$ and $g: \mathbb{R} \to \mathbb{R}$ be the functions.

$$f(x) = \begin{cases} x^2 + 2 & (x \leqslant 0) \\ 2 - x^2 & (x > 0) \end{cases} \quad g(x) = \begin{cases} -\sqrt{x - 2} & (x \geqslant 2) \\ \sqrt{2 - x} & (x < 2) \end{cases}$$

  Prove that $g = f^{-1}$.

  - **Proof 11.1.1**
    We need to prove that $g \circ f = \mathrm{id}_{\mathbb{R}}$, that is, $(g \circ f)(x) = x$ for all $x \in \mathbb{R}$.
    If $x \leqslant 0$ then $x^2 \geqslant 0$ and $x^2 + 2 \geqslant 2$, hence $(g \circ f)(x) = g(f(x)) = g(x^2 + 2) = -\sqrt{(x^2 + 2) - 2} = -\sqrt{x^2} = -|x| = -(-x) = x$.
    If $x > 0$ then $x^2 > 0$ and $2 - x^2 < 2$, hence $(g \circ f)(x) = g(f(x)) = g(2 - x^2) = \sqrt{2 - (2 - x^2)} = \sqrt{x^2} = |x| = x$.

- **Theorem 11.2**
  A function $f: X \to Y$ is invertible, if and only if, f is bijective.

  - **Proof 11.2.1**
    ($\Longrightarrow$) Suppose that f is invertivle then there exists $f^{-1}: Y \to X$ such that $f^{-1} \circ f = \mathrm{id}_X$ and $f \circ f^{-1} = \mathrm{id}_Y$.
    If $f(x) = f(x')$ then

$$\begin{gathered} f^{-1}(f(x)) = f^{-1}(f(x')) \\ (f^{-1} \circ f)(x) = (f^{-1} \circ f)(x') \\ \mathrm{id}_X(x) = \mathrm{id}_X(x') \\ x = x' \end{gathered}$$

    hence f is injective.
    Let $y \in Y$ then $y = \mathrm{id}_Y(y) = (f \circ f^{-1})(y) = f(f^{-1}(y))$ with $f^{-1}(y) \in X$, hence f is surjective. Therefore, f is bijective.

    ($\Longleftarrow$) Suppose that f is bijective. Let $y \in Y$ then $y \in \operatorname{Ran}(f)$ because f is surjective and since f is injective then by the previous theorem there exists a unique $x \in X$ such that $y = f(x)$.

Define $g: Y \to X$ by $g(y) = x$ where x is the unique element in X such that $y = f(x)$ then

$$g \circ f = g(f(x)) = x = \mathrm{id}_X(x)$$
$$f \circ g = f(g(x)) = y = \mathrm{id}_Y(y)$$

therefore $g \circ f = \mathrm{id}_X$, $f \circ g = \mathrm{id}_Y$ and thus f is invertible.

- **Definition 12**
  Let $f: A \to B$ and $g: B \to A$ be two functions.

  g is a **left inverse** of f if $g \circ f = \mathrm{id}_A$.

  g is a **right inverse** of f if $f \circ g = \mathrm{id}_B$.